



Htet Myat Aung (Alexander E. Hong)

✉ htetmyataung.ctla@gmail.com | ☎ +66(0)812900801 | 📍 Bangkok, Thailand | 🔗 [LinkedIn](#) | 🌐 [alexhonghma.github.io](#)

PROFESSIONAL SUMMARY

Automation Engineering graduate with hands-on experience in computer vision, 3D deep learning, and robotic perception systems. **Best Paper Award winner at IEEE ICoICT 2025** for developing an optimized 3D CNN achieving 99.75% accuracy in real-time infrared gas leak detection. Developed SLAM algorithms optimized for edge deployment on embedded platforms. Proficient in Python and C++ for perception applications with practical experience in camera calibration, sensor integration, and ROS2-based autonomous navigation. Strong foundation in image processing, temporal-spatial feature extraction, and algorithm optimization for resource-constrained robotic environments.

RESEARCH & PROFESSIONAL EXPERIENCE

Full Time Machine Learning Researcher

Sep. 2025 – Present

Maintenance Technology Center (MTC) – Non-Destructive Detection Technology Lab, KMUTT

Bangkok, TH

- **Co-authored Journal Publication:** Co-authored “Development of High-Speed Inspection Technique Based on Motion-Induced Eddy Current Testing for Surface Railway Defects,” published in *Nondestructive Testing and Evaluation* (Taylor & Francis, Mar. 2026). Contributed to MIECT probe development and signal processing for detecting cracks as small as 0.5 mm at speeds up to 30 km/h on ferromagnetic rail steel. DOI: [10.1080/10589759.2026.2638558](#)
- **Production Pipeline Engineering:** Architected and maintained scalable ETL pipelines using Python to process high-volume acoustic emission sensor data.
- **Workflow Automation:** Optimized end-to-end ML workflows, reducing data processing latency and accelerating model training lifecycles through automated scripting and rigorous version control.
- **Model Reliability & Scalability:** Engineered robust anomaly detection systems for industrial pipe inspection, focusing on reproducibility and scalable deployment in Linux-based environments.
- **Data Quality Assurance:** Implemented automated validation steps within the data ingestion pipeline to ensure high-fidelity input for unsupervised clustering models.

Computer Vision (Student) Researcher

Jun. 2024 – Sep. 2025

Department of Control Systems and Instrumentations (INC), KMUTT

Bangkok, TH

- Developed optimized 3D CNN architecture for real-time infrared video classification achieving **99.75% accuracy** in methane gas leak detection, processing temporal-spatial features from thermal camera streams.
- Implemented classical image processing pipeline including Gaussian Mixture Models (GMM), Running Average, and adaptive background subtraction for robust foreground segmentation in varying lighting conditions.
- Optimized model for edge deployment through input resolution scaling and quantization techniques, maintaining 99%+ accuracy while **reducing inference time by 60%**.
- **Published Conference Paper:** H. M. Aung and S. Wongsu, “Optimised 3D-CNN for Real-Time Infrared Natural Gas Leak Classification: Balancing Accuracy and Computational Cost,” in *Proc. 13th IEEE International Conference on Information and Communication Technology (ICoICT 2025)*, Bandung, Indonesia, Jul. 2025, pp. 1–6. **Best Paper Award.** DOI: [10.1109/ICoICT66265.2025.11192973](#)

Machine Learning Intern

Jun. 2025 – Aug. 2025

Maintenance Technology Center (MTC) – Non-Destructive Detection Technology Lab, KMUTT

Bangkok, TH

- Independently designed and implemented unsupervised ML solutions for GRP pipe inspection using acoustic emission sensor data on Linux-based WSL2 platform with Git version control.
- Won **First Runner-Up Award** in “Smart Infrastructure for Smart City” competition by Metropolitan Waterworks Authority.
- Listed in **KMUTT Hall of Fame** for outstanding achievement in Smart Infrastructure development.

Junior ML / Large Language Model Intern

Dec. 2024 – Feb. 2025

HoneyHunt.AI

Bangkok, TH

- Designed and deployed agentic workflows using Generative AI to automate data processing and decision-making for AI-powered recruitment, directly supporting operational efficiency.
- Improved algorithm scalability and matching accuracy for production-level deployment.
- Implemented API management best practices including versioning, rate limiting, and authentication using API gateways for reliable GenAI workflow deployment.

Perception (Student) Research Engineer

CODIA Lab, KMUTT & Panasonic Singapore

- Developed and optimized SLAM Sparse Bundle Adjustment algorithms in C for visual odometry applications targeting real-time edge deployment on ARM-based embedded platforms.
- Optimized SLAM Sparse Bundle Adjustment algorithms for edge devices, improving computational efficiency and enabling real-time performance in resource-constrained environments.

Feb. 2024 – Jul. 2024

Bangkok, TH

Autonomy and FormulaStudent Vehicle Control Student Researcher

COSIN Lab, INC Department, KMUTT

- Developed ROS2-based autonomous navigation systems integrating IMU and camera sensor fusion for competitive FormulaStudent robotics applications.
- Implemented micro-ROS framework on microcontrollers for real-time IMU data acquisition and sensor synchronization, achieving sub-10ms latency for control loop feedback.
- Performed sensor calibration including IMU bias estimation and camera extrinsic alignment to improve localization accuracy in dynamic environments.

Nov. 2023 – Feb. 2024

Bangkok, TH

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT)

Bachelor of Engineering in Automation Engineering

- GPAX: 3.66/4.00 (First Class Honors)
- Thesis: Natural Gas Methane Leak Detection Using an Infrared Camera
- Department: Control System and Instrumentation (INC)

Graduated Sep. 2025

Bangkok, Thailand

LEADERSHIP & ACTIVITIES

Founder – Laboratory of Robotics and AI (L.O.R.A)

Robotics Research Laboratory

- Founded and led a comprehensive robotics research laboratory focusing on practical applications.
- Spearheaded development of autonomous disinfection robots during COVID-19 pandemic in Myanmar.
- Managed cross-functional communication between engineering teams, public health stakeholders, and external partners to align technical development with societal needs.

Dec. 2019 – Jun. 2021

Yangon, Myanmar

Mentor – FIRST GLOBAL Challenge 2019

Myanmar National Robotics Team

- Provided technical guidance and strategic mentorship to Myanmar's national robotics team through structured training sessions and collaborative problem-solving.
- Led team to win Judges Award at international robotics competition.

Jun. 2019 – Oct. 2019

Dubai, UAE

AWARDS & RECOGNITION

Best Paper Award – 13th IEEE International Conference on ICT (IcoICT 2025)
First Runner-Up – Smart Infrastructure for Smart City, Metropolitan Waterworks Authority
KMUTT Hall of Fame – Outstanding Achievement in Smart Infrastructure Development
Judges Award – FIRST GLOBAL Challenge 2019 (Team Mentor, Myanmar National Team)

TECHNICAL SKILLS

Perception & Vision: OpenCV, SLAM (Sparse Bundle Adjustment), Computer Vision, Image Processing, Background Subtraction, Feature Extraction, Camera Calibration
Robotics: ROS2, Robot Kinematics & Control, Sensor Fusion, micro-ROS, Autonomous Navigation, Gazebo Simulation
Programming: Python, C++, C & Embedded C, Bash Scripting
ML / DL: PyTorch, TensorFlow, 3D CNN, Object Detection (YOLO), Segmentation, Model Quantization
Tools: Git, Linux, WSL2

LANGUAGES

English: Fluent (TOEFL 84 – Writing: 25, Speaking: 26); extensive technical documentation and international collaboration experience
Burmese: Native